Solafune Sentinel-2 Change Analysis

Detailed Usage Guide — QGIS Plugin Edition & Script (CLI) Edition

This document walks through installation, configuration, and usage — screen by screen, field by field — for both execution modes included in the submission bundle (`submission/`): (1) the QGIS plugin edition, run interactively inside QGIS, and (2) the script (CLI) edition, run from a terminal. It also covers project background, the rationale behind each algorithm, the spatial database schema, bugs actually found and fixed during development, and the test/verification record. Both editions share the exact same analysis engine (the `solafune_change` core package), so results are identical regardless of which one you use.

Item	Value
GitHub repository	https://github.com/yeonjun7724/solafune-sentinel2-change (Public)
AOI	Open-pit mine, Zambia, approx. 264.6 km² (EPSG:32735, UTM 35S, 10 m resolution)
Compared acquisitions	2023-08-12 vs 2023-09-02 (Sentinel-2 bands B02/B03/B04)
Headline results	514 change features / total changed area 37,405,887 m² (14.14% of AOI) / Global Moran's I = 0.834 (p=0.001)
Target QGIS version	3.28 or later (installed and run on a real 3.44.12 instance)
Target Python version	3.10 or later (developed/tested on 3.12)
Tests	65 pytest cases, all passing; ruff/black clean

Table of Contents

0. Submission Bundle Contents (Full Layout)

The `submission/` folder contains everything needed for review in one place: both runnable editions, the actual final outputs, a copy of the raw input data, and this document.

Path	Description	QGIS needed?
Usage_Guide.pdf	This document	-
solafune_change_analyzer.zip (+.sha256)	(1) QGIS plugin install package (analysis engine vendored inside). Checksum file included	Yes
solafune-sentinel2-change-source.zip	(2) Script (CLI) edition, and also the complete source repository as a single archive (wrapped in a solafune-sentinel2-change-master/ folder, matching GitHub's "Download ZIP" layout). Built with git archive from the exact commit pushed to GitHub, excluding only the submission/ and yeonjun/ packaging folders themselves (see the note below)	No
data/	(3) A standalone copy of the raw input satellite imagery (AOI + band GeoTIFFs), for quick inspection	No
results/outputs/	(4) The actual final outputs produced by a real pipeline run — GeoPackage database, static comparison figure, interactive map, spatial-statistics JSON/CSV, report.md, and more. Open directly, no re-run required	No (opens best in QGIS)
results/data_processed/	(4) The stacked GeoTIFF and the baseline/CVA intensity and binary change rasters, as produced	No

Item (2) is the "entire repository as one zip file." It is built directly from the commit actually pushed to GitHub via `git archive`, wrapped in the same `solafune-sentinel2-change-master/` folder GitHub's Code → Download ZIP button produces — the full source code, tests, docs, and configuration, byte-for-byte from that commit. It is at the same time the "script/CLI edition" package: unzip it and it is immediately runnable (see PART B). One deliberate exclusion: the `submission/` and `yeonjun/` packaging folders (this very bundle, and the author's personal working notes) are left out of the archive, since including them would make the zip contain a copy of itself and grow without bound every time it is rebuilt. Everything needed to build and run the project from scratch is included.

Item (4) (results/) is the "final output artifacts." These are copies of what the pipeline actually produced on a real run, placed here so they can be opened immediately without re-running anything — GeoPackage, figures, map, statistics. (Item (2), the source zip, does not include these output files, to keep its size reasonable; open results/ instead if you just want to see the outcome without executing the pipeline.)

1. Original Assignment Requirements vs. Delivered Work

The five parts required by `instructions.pdf`, checked one by one against what was actually delivered.

Part	Requirement	Delivered / verified
1. Data Preparation	Load B2/B3/B4, verify CRS/transform/dimensions, stack bands	Verified via validation.py; produces data/processed/sentinel2_<date>_stack.tif with exactly the expected naming
2. Change Detection	May reference the example, but must not copy it verbatim; own algorithm required	baseline.py (independent reimplementation of the example's idea) plus cva.py (robust CVA, the primary method) — both implemented, each producing intensity and binary rasters
3. Feature Extraction & Storage	Polygonize, store in SQLite or PostGIS, id/date_before/date_after/area_m2/confidence/geometry	Polygonized via vectorization.py, stored in GeoPackage (SQLite-based) with a real geometry column; all required fields present plus 15 additional fields (spatial statistics, ML)
4. Visualization	AOI polygon, change raster/polygon display (interactive is a plus)	Both a static comparison figure (PNG) and a Folium interactive map (HTML, works fully offline) were actually generated and verified
5. Analysis & Interpretation	report.md with Method / Results / Interpretation sections	Expanded to 7 sections including those three, populated with numbers from an actual run

All required deliverables (code, database, README.md, report.md) are present. On top of that, spatial statistics, unsupervised spatial ML, a QGIS plugin, and a 65-case automated test suite were added, none of which the assignment required.

PART A. QGIS Plugin Edition

A.1 Prerequisites

- QGIS 3.28 or later (developed and verified against QGIS 3.44.12, OSGeo4W build)
- Install package: `solafune_change_analyzer.zip`

A.2 Installation Steps

- Launch QGIS
- Menu: Plugins → Manage and Install Plugins → Install from ZIP
- Select `solafune_change_analyzer.zip` → Install Plugin
- In the Installed tab, confirm "Solafune Change Analyzer" is checked
- Open the dock widget via the toolbar icon or Plugins → Solafune Change Analyzer

A.3 Why Dependencies Are Missing

The plugin's analysis engine depends on standard geospatial-science Python packages: rasterio, geopandas, scikit-image, scikit-learn, libpysal, esda, matplotlib, and folium. The Windows/OSGeo4W build of QGIS does not ship these inside its own Python by default (QGIS's own GIS functionality talks to GDAL/PROJ directly at the C++ level, so it has no built-in need for a Python-level rasterio, etc.). Checked against a real QGIS 3.44.12 install:

Package	Bundled with QGIS?	What it's needed for
numpy, geopandas, shapely, pyproj, scipy, PyYAML, matplotlib, folium	Yes (Ready)	Core execution, visualization
rasterio	No (Missing)	Raster I/O — the most critical one; nothing runs without it
scikit-image	No (Missing)	Morphological post-processing
libpysal, esda	No (Missing)	Spatial statistics (Moran's I, Getis-Ord Gi*)
scikit-learn	No (Missing)	Experimental unsupervised spatial ML

A.4 Two Ways to Install the Dependencies

"Do I have to set up a separate `.venv` folder?" Short answer: no, you don't. Both options below were actually field-tested end to end against a real QGIS 3.44.12 install.

Option 1 (recommended, no extra folder): install straight into QGIS's own Python

Run a single `pip install --user` against QGIS's own bundled Python interpreter. After that, the Dependencies tab automatically reports "Ready," and Embedded mode works as-is — no External interpreter configuration needed at all. Packages land under `%APPDATA%\Python\Python312\site-packages` (a per-user folder, no admin rights required), which is already on QGIS's `sys.path`.

Windows (adjust the path if your QGIS is installed elsewhere):

```
cd "C:\Program Files\QGIS 3.44.12\bin"
python-qgis-ltr.bat -m pip install --user rasterio geopandas shapely pyproj scipy scikit-image
PyYAML libpysal esda scikit-learn matplotlib folium
```

Field-verified result (QGIS 3.44.12, after running the command above): every required package in the Dependencies tab flipped to Ready; rasterio coexisted with QGIS's own GDAL in the same process without conflict (CRS lookups etc. worked normally); and both Validate Inputs and a full pipeline run (Run Analysis, with spatial statistics and experimental ML enabled) succeeded start to finish in Embedded mode (514 change features, Global Moran's I = 0.804 — a sane value). This was verified for this specific QGIS-version / package-version combination; other QGIS/OS combinations could in theory still hit a GDAL version conflict (Option 2 below eliminates that risk entirely).

Option 2 (isolated, safer): separate .venv + External interpreter

Option 1 loads a second GDAL into the same process as QGIS, which carries a theoretical conflict risk depending on the QGIS build. If you want full isolation (so nothing you install can affect QGIS itself or other plugins), create an independent `.venv` that runs in a separate process, and point the plugin at it via External interpreter mode. Follow PART B's B.2 installation steps to create the `.venv`, then set that environment's `python.exe` path in the Dock Widget's Dependencies tab.

	Option 1: install into QGIS's own Python	Option 2: separate .venv
Extra folder needed	No	Yes (.venv/)
Execution process	Same process as QGIS (embedded)	Separate process (external call)
Setup steps	One pip command	Create .venv + install + set the path in the Dependencies tab
GDAL conflict risk	Theoretical (none observed on QGIS 3.44.12 in testing)	None (fully isolated separate process)
Speed	Slightly faster (no process-spawn overhead)	Slightly slower (a new process per run)
Recommended for	Wanting something quick and simple	Not wanting to affect other QGIS plugins/projects

A.5 Dock Widget — All 6 Tabs, Every Field

① Inputs tab

Field	Description
Before / After Sentinel-2 folder	Each folder must contain B02/B03/B04 GeoTIFF or JP2 files directly (not in a subfolder). Example: inputs/data/sentinel2_20230812
AOI vector file	GeoJSON, GeoPackage, or Shapefile. Does not need to be in WGS84 — it is automatically reprojected to match the raster CRS
Output directory	Where results are written. Created automatically if it doesn't exist
Run label	Optional label to tell runs apart. Prepended to the run_id and shown in the layer group name / run_metadata. Left blank, only a timestamp is used
Validate Inputs button	Checks CRS, resolution, and band presence, then shows a Valid / Valid with warnings / Invalid badge plus a band metadata table
Clear Inputs (reset) button	(New) Clears all four path fields, the Run label, and the validation display. Also clears the values persisted from the previous session, so stale paths don't linger after restarting QGIS. Does not touch settings in other tabs (Change Detection, etc.)

② Change Detection tab

Field	Description
Method	Provided baseline (example method only) / Robust RGB CVA (improved method only) / Run both (default, runs both and compares)
Radiometric normalization	None (no correction) / Robust median/MAD (default) — linear matching using the median and MAD of valid pixels common to both dates / Percentile matching — matched using the 2nd/98th percentiles / PIF (Pseudo-Invariant Features) — linear regression over pixels with low change only
Threshold method	Otsu (default) — automatic binarization / Percentile — flags anything above a given percentile as change / Manual — an explicit absolute value
Morphology	Whether to apply opening/closing morphological operations to the binary raster, and the kernel size (px)
Minimum change area	Connected components smaller than this are treated as noise and dropped (in m^2; converted automatically from pixel area)

③ Spatial Analysis tab

Field	Description
Enable spatial statistics	Turns on grid-aggregated spatial statistics (on by default)
Grid cell size	Side length of the analysis grid, in meters. Default 150 m — statistics are computed after aggregating to this grid rather than at pixel resolution, to avoid a dense spatial weight matrix
Spatial weights	Queen contiguity (default, shares an edge or a corner) / Rook contiguity (edge only) / K nearest neighbors
Permutations	Number of Monte Carlo permutation-test iterations (default 999, reproducible via a fixed seed)
Significance alpha	Significance level (default 0.05)
FDR correction	Benjamini-Hochberg correction — mitigates the inflated significance that comes from testing many locations at once with Gi*
Global/Local Moran's I	Global: spatial autocorrelation of overall change intensity. Local (LISA): per-cell High-High / Low-Low / High-Low / Low-High cluster classification
Getis-Ord Gi*	Detects statistically significant hot spots / cold spots (90/95/99% bands)
Experimental unsupervised spatial ML	Since there is no ground truth at all, a warning is always shown on screen. Choose Isolation Forest (anomaly score) or DBSCAN (spatial clustering), with tunable hyperparameters (contamination / eps / min_samples, etc.)

④ Outputs tab

- Whether to keep intermediate GeoTIFFs/stacks, whether to build the interactive map, whether to auto-apply QML styling
- Whether to auto-load results into the current QGIS project, and whether to zoom to them afterward
- Whether to keep temporary files from a failed or cancelled run

⑤ Run & Results tab

- Run Analysis — shows a progress bar, current stage, elapsed time, and a live log
- Cancel — safe to stop at any point (atomic file writes ensure a partial output is never mistaken for a completed result)

- On completion, a summary (feature count, total changed area, Global Moran's I, permutation p-value, 95% hotspot count, mean confidence, run time) is shown alongside Open Report / Open Interactive Map / Open Output Folder / Add Results to Map / Copy Summary / Save Log buttons

⑥ Dependencies tab

- Execution environment: Automatic (default — embedded when possible, external otherwise) / force Embedded / force External
- Set an External interpreter path and check its package status live with the Check dependencies button
- A Ready/Missing table per package, including version info
- Restore defaults / Import configuration YAML / Export configuration YAML

A.6 Run Progress (12 Stages)

Progress	Stage
0–5%	Input discovery — locating B02/B03/B04 files
5–12%	Validation — CRS / transform / dimensions / AOI overlap checks
12–22%	Raster alignment — grid alignment (resampling if needed), AOI masking
22–30%	Stack creation — building the RGB-ordered band-stack GeoTIFF
30–42%	Radiometric normalization — applying the chosen correction
42–55%	Baseline detection — example-method change detection
55–68%	Robust CVA — improved-method change detection
68–75%	Threshold & morphology — binarization plus post-processing
75–82%	Polygon extraction — vectorization, confidence scoring
82–90%	Spatial statistics — Moran's I / Gi* / FDR
90–94%	Experimental spatial ML — (if enabled) Isolation Forest / DBSCAN
94–100%	Database, visualization & report — GeoPackage, figures, map, report.md

A.7 Result Layer Tree

```
Solafune Change Analysis - <run id>
Inputs
AOI, Before RGB, After RGB
Change Detection
Baseline Intensity/Binary, CVA Intensity/Binary, Change Features
Spatial Statistics
Analysis Grid, LISA Clusters, Gi* Hotspots
Experimental ML
Spatial Anomalies (only if ml_enabled=True)
```

Each layer gets a pre-built QML style applied automatically (continuous rasters have their color ramp rescaled at runtime to the actual data range). Groups for results that don't exist for a given run (e.g. Spatial Anomalies when ML is disabled) are simply not created.

A.8 Using the Processing Toolbox

The same engine can also be run from Processing Toolbox → Solafune Geospatial Analytics → Sentinel-2 Change Analysis (15 parameters, useful for batch processing and Model Builder integration). It goes through the same request builder as the dock widget and calls the same `run_pipeline()`, so the analysis logic is never duplicated.

A.9 Before/After Comparison

Toggle the visibility checkboxes on the “Before RGB”/“After RGB” layers in the result layer group, or adjust their opacity in the layer panel, to compare the two dates. The RGB stretch is applied consistently across both dates (the 2nd/98th-percentile stretch computed on the before image is reused for the after image), so the visual comparison isn't skewed by independently-stretched images.

A.10 Troubleshooting

Symptom	Cause / fix
An error dialog on “Validate Inputs” (used to be a hard crash)	rasterio, etc. missing from the embedded Python — check status in the Dependencies tab, install via Option 1 or 2 in A.4
Before/After folder selected but no bands are found	B02/B03/B04 files must sit directly inside the folder (not in a subfolder)
Stale paths from a previous session keep reappearing	Use the Clear Inputs button on the Inputs tab to clear both the fields and the persisted values
Results are written to an unexpected folder (e.g. under some unknown temp directory)	This was a bug in an earlier version (processed_dir path computation) — fixed in the current version; the path is now always computed correctly relative to the output directory
Run Analysis doesn't seem to progress	Check the “Solafune Change Analyzer” tab under QGIS's “View → Panels → Log Messages” panel for detailed logs
Result layers don't appear	Check whether “Load results into current QGIS project” is enabled on the Outputs tab
External-mode run never finishes	Confirm the interpreter path points at an actual python.exe (not a .bat file or shortcut), and that pip install -e . has been run in that environment

A.11 Checksum Verification (Optional)

Run this in PowerShell or Command Prompt and compare the printed hash against the contents of `solafune_change_analyzer.zip.sha256`.

```
certutil -hashfile solafune_change_analyzer.zip SHA256
```

PART B. Script (CLI) Edition

B.1 Prerequisites

- Python 3.10 or later (developed/tested on 3.12)
- Install package: `solafune-sentinel2-change-source.zip`

B.2 Installation Steps

1) Unzip (folder name ends in -master, matching GitHub's Download ZIP layout)

```
unzip solafune-sentinel2-change-source.zip
cd solafune-sentinel2-change-master
```

2) Create a virtual environment

```
python -m venv .venv
```

3) Install dependencies

```
.venv\Scripts\pip install --upgrade pip
.venv\Scripts\pip install -r requirements-dev.txt
.venv\Scripts\pip install -e .
```

On macOS/Linux, replace `.venv\Scripts\pip` with `source .venv/bin/activate` followed by `pip ...`.

B.3 One-Line Run (Recommended)

```
solafune-change all --config config/default.yaml
```

The full pipeline finishes in roughly 30 seconds; results land in `outputs/` and `data/processed/`.

B.4 Individual CLI Commands

Command	Description
solafune-change validate --config <yaml>	Validates inputs only (dry run)
solafune-change run --config <yaml>	Runs the full pipeline
solafune-change stats --config <yaml>	Forces spatial statistics on
solafune-change report --config <yaml>	Regenerates report.md from a prior run
solafune-change all --config <yaml>	validate then run (recommended)
python -m solafune_change --help	Same as above, invoked as a module

B.5 Full Option Reference

Option	Description
--before / --after / --aoi / --output-dir	Path overrides
--method baseline|cva|both	Analysis method (default both)
--threshold-method otsu|percentile|manual	Threshold method
--percentile N	Percentile for the percentile method
--threshold-value N	Absolute value for the manual method

Option	Description
`--min-area-m2 N`	Minimum change-feature area
`--grid-size-m N`	Spatial-statistics grid size
`--permutations N`	Number of permutations
`--seed N`	Random seed
`--ml / --no-ml`	Toggle experimental ML on/off
`--json-progress`	JSON Lines progress output (used for external/plugin execution)
`-v / --verbose`	Debug-level logging

Example

```
solafune-change run --config config/default.yaml \
--threshold-method percentile --percentile 97 \
--min-area-m2 900 --grid-size-m 200 --seed 7 --ml
```

B.6 Output File Locations

Path	Contents
data/processed/sentinel2_<date>_stack.tif	RGB-ordered band stack
data/processed/{baseline,cva}_change_{intensity,binary}.tif	Four change-detection rasters
outputs/database/change_analysis.gpkg	GeoPackage (SQLite-based spatial database)
outputs/figures/change_comparison.png	Baseline vs. CVA vs. Gi* comparison figure
outputs/maps/interactive_map.html	Interactive map (fully offline)
outputs/statistics/global_moran.json, spatial_statistics.csv	Spatial-statistics values
outputs/qgis/styles/*.qml, solafune_change_analyzer.zip	QGIS styles, plugin zip
outputs/{summary,run_manifest,quality_report}.json, report.md	Summary, metadata, and report

B.7 Rebuilding the QGIS Plugin ZIP

```
python scripts/build_qgis_plugin.py
python scripts/validate_qgis_plugin.py outputs/qgis/solafune_change_analyzer.zip
```

B.8 Regenerating This PDF

This PDF is itself generated by a script (a reportlab script that embeds the Malgun Gothic font, which also renders Latin text cleanly).

```
python scripts/generate_submission_pdf.py --out submission/Usage_Guide.pdf
```

B.9 Tests / Code Quality

```
python -m pytest tests/ -v
ruff check .
black --check src/ tests/ scripts/ qgis_plugin/solafune_change_analyzer --exclude vendor
```

B.10 Full config/default.yaml Field Reference

Section.field	Default	Description
paths.aoi/before_folder/after_folder	-	Required input paths
paths.output_dir	outputs	Output directory
paths.processed_dir	(auto-computed from output_dir if unset)	Intermediate GeoTIFF folder
preprocessing.normalization	robust_median_mad	none / robust_median_mad / percentile_matching / pif_linear
preprocessing.reflectance_scale	10000.0	DN-to-reflectance scale factor
change_detection.method	both	baseline / cva / both
change_detection.threshold_method	otsu	otsu / percentile / manual
change_detection.min_area_m2	400.0	Minimum change-feature area
change_detection.morphology.*	opening_then_closing, 3px	Morphological post-processing
spatial_statistics.enabled	true	Toggle spatial statistics
spatial_statistics.grid_size_m	150.0	Analysis grid size
spatial_statistics.weights	queen	queen / rook / knn
spatial_statistics.permutations	999	Number of permutations
spatial_statistics.fdr_correction	true	Toggle BH-FDR correction
spatial_ml.enabled	false	Toggle experimental ML (opt-in)
spatial_ml.model	isolation_forest	isolation_forest / dbscan
run.random_seed	42	Global reproducibility seed

B.11 Troubleshooting

Symptom	Cause / fix
"command not found: solafune-change"	Virtual environment not activated, or pip install -e . was not run
CRS/band errors on the input files	Diagnose first with solafune-change validate --config ...
Run is slow or hangs	Retry with a larger grid-size-m or fewer permutations
Result numbers differ from report.md	Confirm you re-ran with the same seed and config (reproducibility is guaranteed when they match)

2. Actual Analysis Results (config/default.yaml, seed=42)

Metric	Value
Change feature count	514
Total changed area	37,405,887 m² (14.14% of AOI)
Primary method / threshold	Robust CVA / Otsu (value = 5.2199)
Vs. baseline	baseline flags 47.85% of pixels as changed (far noisier than CVA's 15.32%)
Global Moran's I	0.8335 (E[I]=-0.0001, z=178.37, permutation p=0.001, 999 permutations, seed=42)
Features overlapping a 95%+ Gi* hotspot	95
Largest change feature	15,987,700 m²
Experimental ML (Isolation Forest, opt-in)	Top-K outlier stability (spatial block bootstrap) = 0.72
Run time	~30 seconds (full pipeline, CLI)

Interpretation: the strong, statistically significant spatial clustering of change (Moran's I = 0.834, p = 0.001) is consistent with genuine, structured surface change rather than noise — clustering is exactly what real, connected surface disturbance (excavation, waste-rock placement, land clearing, vegetation removal) would produce, and exactly what random sensor or atmospheric noise would not. However, two RGB-only dates cannot by themselves confirm which of those specific activities is responsible, and cannot rule out non-mining confounds. Both report.md and this guide deliberately use "consistent with" phrasing and stop short of any definitive land-use classification.

Interpreting uncertainty: real change vs. clouds, shadow, season, brightness

Each alternative explanation for a detection is addressed explicitly below, rather than assumed away:

Possible confound	Could it explain a detection here?	How this pipeline addresses it
Clouds / cloud shadow	Not ruled out — no Scene Classification Layer (SCL) or cloud mask was supplied	Not filtered; explicitly flagged as an unresolved limitation, never silently ignored
Terrain / pit-wall shadow	Possible at steep pit edges where illumination geometry differs slightly between dates	Robust median/MAD normalization plus Otsu thresholding reduce (not eliminate) shadow-driven false positives; morphological opening removes single-pixel shadow speckle
Seasonal change (vegetation, soil moisture)	Low risk for this pair (~3 weeks apart), but not verified in general	Cannot be separated from persistent change with only two dates; a denser time series is recommended in report.md's Operational Recommendations
Whole-scene brightness / illumination difference	Likely present to some degree between any two acquisitions	Directly targeted by the robust_median_mad radiometric-normalization step before CVA is computed — this is precisely why baseline (no normalization) is far noisier than CVA
Small co-registration offset	Possible at sharp edges (pit rim, road margins)	Not explicitly corrected; the minimum-change-area filter (400 m²) removes some of the resulting small spurious polygons

None of these confounds can be fully excluded with two RGB-only dates and no cloud mask. This table is reproduced verbatim in `report.md` Section 5 (Interpretation), so it travels with every run, not just this document.

3. Algorithm Choices and Rationale

Baseline (reimplementation of the provided example)

Computes per-pixel multi-band Euclidean distance and min-max normalizes it. Because it applies no radiometric correction and uses global min-max scaling, a single extreme-value pixel can distort the entire intensity scale — this weakness was deliberately preserved by reimplementing the example's core idea faithfully (not copied; an independent implementation with type hints, logging, and error handling).

Robust RGB CVA (primary method)

Per-band differences are standardized by median/MAD before being combined: `z_b = (diff_b - median(diff_b)) / (1.4826 * MAD(diff_b))`, `CVA = sqrt(z_B04^2 + z_B03^2 + z_B02^2)`. If MAD is close to zero, the standard deviation is used as a safe fallback. In practice, CVA flags only 15.32% of pixels as changed versus baseline's 47.85%, confirming it is markedly less sensitive to noise.

Spatial statistics

Rather than building a dense pixel-level spatial weight matrix, values are first aggregated to a 150 m grid (11,967 cells), then Global/Local Moran's I is computed with Queen contiguity weights and Getis-Ord Gi* with binary weights. Gi* p-values are corrected with the Benjamini-Hochberg FDR procedure.

Experimental unsupervised spatial ML

With no ground truth available, accuracy/precision/recall are never claimed. Instead, a spatial block bootstrap is used to diagnose whether the top-ranked outliers stay stable under resampling. It is disabled by default (opt-in), and every surface — UI and documentation alike — explicitly labels it as “exploratory.”

Why GeoPackage as the database

The assignment's SQLite requirement is satisfied by GeoPackage, an OGC standard that is itself a SQLite file — openable with no server, and backed by a real geometry column and spatial index rather than WKT text.

4. Full Spatial Database Schema

change_features (514 rows, MULTIPOLYGON, EPSG:32735)

Field	Description
`id`	Feature ID
`date_before / date_after`	Compared acquisition dates (20230812 / 20230902)
`method / threshold_method / threshold_value`	Method and threshold used
`area_m2 / perimeter_m / compactness`	Area / perimeter / Polsby-Popper compactness
`mean_change / max_change / p95_change`	Change-intensity statistics within the feature
`confidence`	A 0-1 heuristic score (not a calibrated probability) — weighted combination of threshold-exceedance magnitude, consistency, size, hotspot significance, and ML rank
`gi_zscore / gi_pvalue / gi_qvalue / hotspot_class`	Getis-Ord Gi* results (q is the FDR-corrected value)
`lisa_cluster`	Local Moran's I cluster type (e.g. High-High)
`ml_anomaly_score / ml_cluster_id`	(If ML enabled) outlier score / cluster ID
`geom`	The actual geometry column (MULTIPOLYGON)

spatial_grid (11,967 cells, POLYGON)

Per-cell mean/median/p90/p95 CVA, changed_proportion, mean per-band difference, local_std_cva, plus the same Gi*/LISA/ML fields found on change_features.

run_metadata (1 row per run)

run_id, timestamp, input paths, CRS/resolution, method/normalization/threshold parameters, spatial_statistics/spatial_ml enabled flags, package_version, random_seed — everything needed to reproduce the run.

quality_checks

Warnings/errors raised during validation (zero for this run).

5. Bugs Actually Found and Fixed During Development

These are problems actually reproduced through running, testing, or real usage — not hypothetical issues found by code review alone.

Bug	How it was found	Fix
QGIS "Validate Inputs" crash	Clicked after installing on a real QGIS instance → rasterio missing, ModuleNotFoundError propagated uncaught	Check dependencies up front and fall back to the External interpreter automatically, or show a friendly dialog otherwise
processed_dir path miscomputed	Reported that results were being written under an unrelated temp folder during real usage → traced the root cause	Now always computed as an absolute path relative to output_dir; two regression tests added
write_intermediate setting silently ignored	Found during a full code review	Now actually controls whether intermediate GeoTIFFs are saved
run_label value was discarded	Found in the same review	Now sanitized into run_id and shown in the layer group name / run_metadata
Intermittent matplotlib TclError	Test suite occasionally failed at random — initially misdiagnosed as a PROJ issue, traced via the actual traceback	Forced the non-interactive Agg backend before importing pyplot (no recurrence over 5 consecutive runs)
Crash when polygonization yields zero features	Found while testing the extreme case where min_area filters out every feature	Empty GeoDataFrame now created with an explicit schema
Garbled Korean text in PDF code blocks	Found while reviewing an earlier draft of this document	Courier-styled code blocks are ASCII-only; localized text was moved to separate paragraphs/tables
Possible unguarded CRS lookup failure at runtime	Found while reproducing a PROJ conflict in the development environment	Added defensive handling so a PROJ lookup failure can't propagate uncaught

6. Test and Verification Record

- All 65 pytest cases pass — synthetic-data unit tests + real-data integration tests + 2 processed_dir regression tests
- ruff check ., black --check clean
- Verified against a real QGIS 3.44.12 install: plugin lifecycle, no duplicate registration on reload, Processing provider registration/teardown, Dependencies tab checked live, the on_validate crash reproduced and re-verified as fixed, Clear Inputs button behavior, and a full pipeline run succeeding via the no-venv (embedded) install path
- Real-data pipeline run: GeoTIFF/GeoPackage re-opened and checked after the run, all 9 queries in docs/example_queries.sql executed successfully against the real database
- Reproducibility: running twice with the same seed produces numerically identical results

7. Key Assumptions and Limitations

- Assumes a Sentinel-2 reflectance scale factor of 10000 (no metadata was provided; standard ESA convention applied)

- Band stack order is R-G-B (B04, B03, B02)
- No B08 (NIR) band, so NDVI cannot be computed
- No cloud/shadow mask (SCL) — cloud contamination cannot be ruled out
- Only two acquisition dates, so seasonal effects cannot be separated from permanent change
- No ground truth — accuracy/precision/recall are never claimed anywhere
- confidence is an explicit heuristic score, not a calibrated probability
- Gi*/Moran's I significance describes spatial pattern only; it does not establish a cause (e.g. mining activity)

Appendix A. Example SQL Queries

`docs/example_queries.sql` contains 9 queries, all executed against the real database as part of verification. A representative example:

```
-- Top 10 change features by area
SELECT id, area_m2, mean_change, confidence, hotspot_class
FROM change_features ORDER BY area_m2 DESC LIMIT 10;

-- Change features intersecting a 95%+ significance hotspot
SELECT id, area_m2, gi_zscore, gi_qvalue
FROM change_features WHERE hotspot_class IN ('hot_95','hot_99');
```

Appendix B. Glossary

Term	Description
CVA	Change Vector Analysis — combines multi-band change into a single vector magnitude
MAD	Median Absolute Deviation — an outlier-robust measure of dispersion
Moran's I	Global spatial-autocorrelation index (-1 to 1; positive = clustered, negative = dispersed)
LISA	Local Indicators of Spatial Association — cluster classification based on Local Moran's I
Getis-Ord Gi*	Statistical significance test for local hot/cold spots
FDR	False Discovery Rate — controls the false-positive rate under multiple testing
confidence	A 0-1 heuristic score, not a calibrated probability
GeoPackage	An OGC-standard spatial data format, internally a SQLite file

---

The GitHub repository's top-level `README.md` and `report.md` also document this project in English.